

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
sasuhorukov@sasuhorukov:~$ ls /etc/ > file.txt
sasuhorukov@sasuhorukov:~$ ls >> file.txt
sasuhorukov@sasuhorukov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
```

Рисунок 1: Запись в файл

```
sasuhorukov@sasuhorukov:~$ grep .conf file.txt > conf.txt
sasuhorukov@sasuhorukov:~$
^[[200~cat conf.txt
^[[201~sasuhorukov@sasuhorucat conf.txt nf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
```

Рисунок 2: Поиск расширения .conf


```
personal/stage02/report/_resources/csl
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-
personal/stage02/report/bib/cite.bib
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-
personal/stage03/report/_resources/csl
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-
personal/stage03/report/bib/cite.bib
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-
personal/stage04/report/_resources/csl
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-
personal/stage04/report/bib/cite.bib
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-
personal/stage05/report/_resources/csl
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-
personal/stage05/report/bib/cite.bib
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-
personal/stage06/report/_resources/csl
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-
personal/stage06/report/bib/cite.bib
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presenta
tion/report/_resources/csl
/home/sasuhorukov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presenta
tion/report/bib/cite.bib
/home/sasuhorukov/git-extended/.git/hooks/commit-msg.sample
/home/sasuhorukov/git-extended/.git/objects/c5
/home/sasuhorukov/git-extended/.git/config
/home/sasuhorukov/conf.txt
sasuhorukov@sasuhorukov:~$
```

Рисунок 3: Поиск файлов

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
sasuhorukov@sasuhorukov:~$  
sasuhorukov@sasuhorukov:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
find: '/etc/credstore.encrypted': Permission denied  
find: '/etc/credstore': Permission denied  
find: '/etc/cups/ssl': Permission denied  
sasuhorukov@sasuhorukov:~$  
sasuhorukov@sasuhorukov:~$  
sasuhorukov@sasuhorukov:~$ find ~ -name "log*" > logfile &  
[1] 6463  
sasuhorukov@sasuhorukov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
sasuhorukov@sasuhorukov:~$ rm logfile  
sasuhorukov@sasuhorukov:~$
```

Рисунок 5: Фоновый запуск процесса

```
sasuhorukov@sasuhorukov:~$  
sasuhorukov@sasuhorukov:~$ gedit &  
[1] 6501  
sasuhorukov@sasuhorukov:~$  
^[[200~ps | grep gedit  
^[[201~sasuhorukov@sasuhorups | grep geditp gedit  
    6501 pts/2    00:00:00 gedit  
sasuhorukov@sasuhorukov:~$  
sasuhorukov@sasuhorukov:~$ kill 6501  
[1]+  Terminated                  gedit  
sasuhorukov@sasuhorukov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df. df displays the amount of space available on the file system containing each file name argument. If no file name is given, the space available on all currently mounted file systems is shown. Space is shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the space available on that file system rather than on the file system containing the device node. This version of df cannot show the space available on unmounted file systems, because on most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is
        usually smaller, it may be larger due to holes in ('sparse') files, internal
        fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units
        of 1,048,576 bytes; see SIZE format below

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
sasuhorukov@sasuhorukov:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89104944 65423552  58% /
devtmpfs        4011280      0    4011280   0% /dev
tmpfs           4035656      92    4035564   1% /dev/shm
tmpfs           1614264     1948    1612316   1% /run
tmpfs            1024         0        1024   0% /run/credentials/systemd-journald.service
tmpfs           4035656     1612    4034044   1% /tmp
/dev/nvme0n1p3 155186176 89104944 65423552  58% /home
/dev/nvme0n1p3 1992552    518800    1352512  28% /boot
/dev/loop1      115712    115712         0 100% /var/lib/napd/snap/hugo/25676
/dev/loop0       68480     68480         0 100% /var/lib/napd/snap/core24/1349
/dev/loop2       49280     49280         0 100% /var/lib/napd/snap/snapd/25935
tmpfs            1024         0        1024   0% /run/credentials/systemd-resolved.service
tmpfs           807128      180    806948   1% /run/user/1014
/dev/sr0        2677920   2677920         0 100% /run/media/sasuhorukov/Fedora-WS-Live-43
sasuhorukov@sasuhorukov:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/95
4      ./git-extended/.git/objects/30
4      ./git-extended/.git/objects/fc
4      ./git-extended/.git/objects/49
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
216    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
813696 .
sasuhorukov@sasuhorukov:~$
```

Рисунок 10: Запуск команды du


```
/home/sasuhorukov/git-extended/.git/objects/ab
/home/sasuhorukov/git-extended/.git/objects/04
/home/sasuhorukov/git-extended/.git/objects/38
/home/sasuhorukov/git-extended/.git/objects/c5
/home/sasuhorukov/git-extended/.git/objects/f8
/home/sasuhorukov/git-extended/.git/objects/a6
/home/sasuhorukov/git-extended/.git/objects/6b
/home/sasuhorukov/git-extended/.git/objects/0c
/home/sasuhorukov/git-extended/.git/objects/95
/home/sasuhorukov/git-extended/.git/objects/30
/home/sasuhorukov/git-extended/.git/objects/fc
/home/sasuhorukov/git-extended/.git/objects/49
/home/sasuhorukov/git-extended/.git/logs
/home/sasuhorukov/git-extended/.git/logs/refs
/home/sasuhorukov/git-extended/.git/logs/refs/heads
/home/sasuhorukov/git-extended/.git/logs/refs/remotes
/home/sasuhorukov/git-extended/.git/logs/refs/remotes/origin
/home/sasuhorukov/monthly
/home/sasuhorukov/reports
/home/sasuhorukov/reports/monthly
/home/sasuhorukov/reports/monthly/monthly
/home/sasuhorukov/ski.places
/home/sasuhorukov/ski.places/equipment
/home/sasuhorukov/ski.places/plans
/home/sasuhorukov/australia
/home/sasuhorukov/play
/home/sasuhorukov/play/games
/home/sasuhorukov/play/games/play
sasuhorukov@sasuhorukov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.